What Makes a Song Popular?

PBA Fall 2026 · Group 13 · Milestone 2 Proposal · Repo: github.com/MichaelCHM0910/pba-project-group-13
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1. Research question

Which audio features of a song (danceability, energy, loudness, length, and others) predict its popularity on Spotify, after accounting for its genre and release year?

The question is **predictive**, with a descriptive first part (how popularity differs across genres and years). We do not claim causal effects: songs are not randomly assigned their features, and artist fame is not in the data.

2. Hypotheses and reasons

- **H1. Danceable, loud, and shorter songs are more popular.** Streaming pays per play, and a play counts after 30 seconds, so shorter songs get replayed more and are skipped less. Danceable, loud songs fit the party, workout, and "hits" playlists that drive most listening.
- **H2. Instrumental songs are less popular.** Vocals and lyrics are what listeners remember, sing, and share.
- **H3. Genre and release year matter more than audio features.** Spotify's score is based on recent plays, so new releases score higher, and some genres (pop) have larger audiences than others (EDM).

3. Data description

Source. "Spotify Songs", TidyTuesday, 21 Jan 2020 (github.com/rfordatascience/tidytuesday, data/2020/2020-01-21). Kaylin Pavlik collected it in January 2020 from the official Spotify Web API with the `spotifyr` R package, by pulling songs from Spotify playlists in six genres. The audio features are Spotify's own measures. A codebook comes with the data (copied to `data/codebook.md`).

Unit of observation: one song. **Period:** popularity is a snapshot from January 2020; release years run from 1957 to 2020 (median 2016). **Observations:** 32,833 song–playlist rows; 28,356 unique songs after removing duplicates. No missing values in the variables we use. Six genres with 4,100–5,400 songs each.

Variable	Meaning	Unit	Role
track_popularity	Spotify popularity, based mostly on recent number of plays	0–100	Outcome (Y)
danceability	Suitability for dancing (tempo, rhythm stability, beat strength)	0–1	X
energy	Perceived intensity and activity	0–1	X
loudness	Average loudness of the track	dB (−60 to 0)	X
duration_min	Song length	minutes	X
instrumentalness	Likelihood the track has no vocals	0–1	X
valence	Musical positiveness (happy vs. sad)	0–1	X
acousticness, speechiness, liveness	Acoustic sound; spoken words; live recording	0–1	X
tempo	Estimated tempo	beats per minute	X
mode	Major or minor key	major / minor	X
genre	Genre of the source playlist	EDM, Latin, pop, R&B, rap, rock	Control
release_year	Album release year	year	Control

4. Key variables and measurement

Outcome: `track_popularity` (0–100), computed by Spotify from how many plays a song has had and how recent they are. We also plan a binary version, "hit" = popularity ≥ 70. **Explanatory variables:** Spotify's audio features, computed by Spotify's audio analysis of each recording. **Controls:** genre and release year.

5. What would count as evidence

- **Supports H1/H2:** in a regression with genre and release year held fixed, danceability and loudness have positive coefficients, duration and instrumentalness negative ones; and adding audio features lowers prediction error (RMSE) on a held-out test set compared with a genre + year model.
- **Contradicts:** coefficients near zero or of the opposite sign, or no gain in test-set accuracy from audio features. That would mean popularity depends on things outside the sound itself (artist, marketing, playlists).
- **First check (done):** correlations with popularity are weak: duration −0.14, instrumentalness −0.12, energy −0.10, danceability +0.05. Mean popularity differs by genre (pop 45.9 vs. EDM 30.7). This already hints that H3 may hold, and energy runs against H1.

6. Plan and limitations

Methods: descriptive plots by genre and year; linear regression with controls; train/test split to compare prediction models; logistic regression for "hit". **Robustness:** repeat without the 9% of songs with popularity 0 (often duplicate re-releases), and within each genre. **Limitations:** songs come from playlists, not a random sample of all music; popularity is one snapshot in time; artist fame is missing, so a famous artist's weak song can still score high. The data contain no personal information.